

Photoswitchable mGluR6 agonist elicits light responses in human retinal organoids through selective recruitment of ON-pathway single units

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Background: Retinal degenerative diseases cause photoreceptor loss and vision impairment. Photopharmacology for vision restoration aims to bypass photoreceptors by directly activating downstream circuits. Prosthe6-20 (p6-20), a photoswitchable mGluR6 agonist, targets ON-bipolar cells to restore light-driven output independently of photoreceptors. P6-20 has demonstrated visual acuity restoration in blindness models both in zebrafish larvae and mice, supporting translation to human systems. Human iPSC-derived retinal organoids (ROs) provide a physiologically relevant ex vivo platform to evaluate such strategies. Current RO-multi-electrode array (MEA) systems face technical challenges including incomplete 9-cis retinal recycling and limited maturation, showing heterogeneous light responses.

Methods: Healthy iPSC derived ROs were plated on MEA for electrophysiological recordings (DIV 210+). Standardized 2-minute dark/light cycles were applied at baseline and after p6-20 (32 μ M). Activity was recorded from 8 electrodes with consistent firing (5 organoids, >0.3 Hz). Light responses were quantified via normalized Light Response Index (LRI). Spike sorting resolved putative single units (≥ 180 spikes) to assess functional heterogeneity masked by population averaging.

Results: At baseline, ROs displayed spontaneous activity but minimal light responses (LRI 0.00 ± 0.06), with no electrodes meeting ON-response criterion (LRI >0.3). With p6-20, electrode-level LRI increased (0.13 ± 0.08), with 1/8 electrodes reaching ON-response threshold. Spike sorting resolved 13 active units at baseline and 19 post-treatment. Critically, single-unit analysis revealed ON-type responses in 3/19 units post-treatment (LRI 0.42-0.85) across 3 electrodes, compared to 1/13 units at baseline. Spike sorting unmasks functional heterogeneity invisible at population level, revealing selective recruitment of mGluR6-expressing neuronal subpopulations.

Conclusions: These findings provide proof-of-concept that p6-20 elicits selective ON-pathway responses in human ROs, in line with previous in vivo results, supporting mGluR6 agonism as a photoreceptor-bypass strategy. The sparse yet specific response pattern highlights both mGluR6 selectivity and need for technical refinement of RO-MEA systems. Single-unit analysis was essential for detecting responses. This establishes a foundation for systematic optimization of RO-MEA platforms with spike sorting for preclinical evaluation of vision restoration therapies.